Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.15	20.10	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.01	-0.2%
power (W), mean	630	642	+1.9%
power (W), peak	651	681	+4.6%
energy (Wh)	212	215	+1.7%
energy, parked workers still on at idle power (Wh)	212	215	+1.7%
CPU used (cores), mean	0.912	1.23	+35.5%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.051	+35.5%
energy per core-hour (Wh)	691	520	-24.8%
node-hours per core-hour	6.58	4.86	-26.2%
pending pods, pod-minutes	0.250	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.83	9.45	+7.0%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.767	0.533	-30.4%
requests timed	1,040	1,095	+5.3%
time over the response line (% of samples)	0.769	0.000	-100.0%
response time (ms), mean	154	96.24	-37.5%
response time (ms), median	127	82.90	-34.5%
response time (ms), 95th percentile	319	157	-50.8%
response time (ms), 99th percentile	470	201	-57.2%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2005	5.3272	-26.0%	[-3.5157, -0.1843]	GAIN
kwh_per_core_hour	0.7522	0.5663	-24.7%	[-0.3454, -0.0230]	GAIN
utilisation	0.0375	0.0525	+40.0%	[+0.0019, +0.0292]	GAIN
pending_pod_minutes_per_hour	0.7619	0.0000	-100.0%	[-2.2857, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2381	1.6466	-26.4%	[-4.3905, +2.5778]	EVEN, no significant difference

What Omni-Compass did

• decisions: 80 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	621	5.00	6.00	0.000	0.000	0.589	0.646
1	6.00	6.00	618	622	5.00	7.50	0.000	0.000	0.594	0.716
2	6.00	6.00	617	624	5.00	8.00	0.000	0.000	0.556	0.755
3	6.00	6.00	625	638	5.00	8.00	0.000	0.000	0.780	1.15
4	6.00	6.00	634	646	8.50	10.00	0.000	0.000	1.02	1.32
5	6.00	6.00	634	645	9.00	10.00	0.000	0.000	1.02	1.33
6	6.00	6.00	638	650	9.75	10.00	0.250	0.000	1.08	1.42
7	6.00	6.00	647	667	10.00	10.00	0.000	0.000	1.37	1.96
8	6.00	6.00	650	674	10.00	10.00	0.000	0.000	1.43	2.09
9	6.00	6.00	647	678	10.00	10.00	0.000	0.000	1.36	2.18
10	6.00	6.00	640	669	10.00	10.00	0.000	0.000	1.20	1.99
11	6.00	6.00	618	624	10.00	10.00	0.000	0.000	0.570	0.780
12	6.00	6.00	619	624	10.00	10.00	0.000	0.000	0.654	0.765
13	6.00	6.00	625	635	10.00	10.00	0.000	0.000	0.782	1.03
14	6.00	6.00	633	650	10.00	10.00	0.000	0.000	0.991	1.42
15	6.00	6.00	635	646	10.00	10.00	0.000	0.000	1.05	1.35
16	6.00	6.00	633	651	10.00	10.00	0.000	0.000	1.01	1.45
17	6.00	6.00	630	636	10.00	10.00	0.000	0.000	0.922	1.08
18	6.00	6.00	617	621	10.00	10.00	0.000	0.000	0.594	0.680
19	6.00	6.00	619	622	10.00	10.00	0.000	0.000	0.627	0.710

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.